

MOHAMED HOSSAM

Backend Developer (Intern / Junior Backend Developer)

Beni Suef, Egypt
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Portfolio: <https://mohossam.netlify.app>
GitHub: <https://github.com/mo-hossam-stack>
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SUMMARY

Backend Developer specializing in building secure, scalable, and production-ready web applications using Django and PostgreSQL. Strong background in system design, authentication flows, database optimization, and backend architecture. Experienced in delivering real-world backend systems that handle concurrent users, media delivery, and deployment constraints. Seeking a Backend Intern or Junior Backend Developer role.

TECHNICAL SKILLS

Backend & Architecture: Django, Django REST Framework, System Design, Service Layer Architecture, Authentication & Authorization (Session-based, JWT/OAuth2), RESTful APIs, Secure Backend Design

Databases & Performance: PostgreSQL, MySQL, MongoDB, SQLite, Database Modeling, Indexing, Query Optimization, Schema Design

Security & Reliability: CSRF/XSS Protection, SQL Injection Prevention, Role-Based Access Control, Secure Media Access (Signed URLs)

DevOps & Tools: Docker, Gunicorn, Nginx, Linux CLI, Git/GitHub, Bash, Postman, CI/CD Basics

Languages: Python, JavaScript (ES6+), C++

PROJECTS

CourseHub – Scalable Course Platform

Django, PostgreSQL, HTMX, Cloudinary | 2025 – Present

GitHub: <https://github.com/mo-hossam-stack/coursehub-django>

- Designed and developed a production-ready course platform capable of serving 10,000+ students with secure, session-based access control.
- Built a layered backend architecture (Views → Service Layer → ORM) to isolate business logic, improve maintainability, and support future scalability.
- Implemented passwordless email authentication using UUID-based verification tokens, attempt limits, expiration handling, and secure session persistence.
- Designed a normalized PostgreSQL schema with indexed lookup fields (public_id, foreign keys) to optimize read-heavy traffic and concurrent access.
- Optimized database queries through controlled service-layer access, deterministic ordering, and filtered queriesets to reduce unnecessary load.
- Integrated Cloudinary CDN for scalable media delivery, including private video streaming with signed URLs, adaptive bitrate streaming, and automatic image optimization (WebP/AVIF).
- Improved frontend-backend efficiency using HTMX for partial page updates, reducing payload size and avoiding SPA overhead.
- Containerized the application using Docker, deployed with Gunicorn behind Nginx, and prepared the system for cloud and horizontal scaling.
- Applied production security best practices including CSRF protection, HTTP-only sessions, secure environment variables, and restricted media access.

Dental Practice Management System (DPMS)

Django, PostgreSQL

- Improved patient data handling and appointment scheduling, reducing manual errors by ~30%.
- Implemented role-based access control and authentication to secure sensitive healthcare data.
- Designed a scalable relational database schema to support daily operational workflows.

Django CRM Web Application

Django, Bootstrap

- Streamlined customer management workflows to improve operational efficiency.
- Implemented secure authentication and modular architecture with reusable templates.

Chess Game

Python, Pygame, NumPy

- Developed a complete chess engine with legal move validation and game state management.
- Designed the system for future scalability (AI opponent, online multiplayer).

EDUCATION

B.Sc. Computer Science – Nahda University, Beni Suef

Expected Graduation: 2028 | GPA: 3.23 / 4.0

CERTIFICATIONS

CS50x – Introduction to Computer Science (Harvard)
ALX Professional Foundations
ALX Backend Web Development Program

ADDITIONAL INFORMATION

Languages: Arabic (Native), English (Fluent – B2)
Open Source: Preparing contributions for Google Summer of Code (GSoC 2026)
Focus: Backend Engineering, System Design, Scalable Web Systems